

TEST RESULTS

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Selected Project: Bus Route & Ticketing Management System

This document records the results of the test cases listed in the final Test Plan. Each Test ID below matches the Test Plan.

Class Being Tested: Route

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
R-TC1	Route stores the supplied values correctly.	Route 3 was created with origin Route 66, destination District 9, distance 12.5 km and fare 30.0.	Pass	Values were shown after creation and saved correctly.
R-TC2	Readable route details are displayed.	The route was displayed in readable form with ID, origin, destination, distance and fare.	Pass	Observed through the route menu.
R-TC3	Decimal distance is stored correctly.	Distance 12.5 km was stored and displayed as 12.5.	Pass	Decimal value remained unchanged.
R-TC4	Base fare is stored correctly.	Base fare 30 was stored as 30.0 and displayed correctly.	Pass	Stored route data matched the input.
R-TC5	Route text is stored because it contains alphabetic characters.	Route 66 and District 9 were accepted and stored.	Pass	Text containing letters and numbers is allowed.

Class Being Tested: RouteManager

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
RM-TC1	Route is created and stored.	The valid route was added successfully and saved.	Pass	Creation and saved state matched the expected result.
RM-TC2	Route is rejected with a clear message.	Empty origin produced: Invalid origin: origin cannot be empty.	Pass	No route was added.
RM-TC3	Route is rejected.	Numeric-only destination produced: Invalid destination: must contain at least one letter.	Pass	No route was added.
RM-TC4	Route is rejected.	Negative distance produced: Invalid distance: distance must be greater than 0.	Pass	No route was added.
RM-TC5	Deletion is prevented.	The system printed that the route cannot be deleted because it has existing trips.	Pass	Referenced route remained unchanged.

Class Being Tested: Trip

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
TR-TC1	Trip stores the supplied values correctly.	Trip 3 was created for route 1 on 2026-12-17 from 08:00 to 09:00 with 4 seats.	Pass	Values matched the supplied input.
TR-TC2	Readable trip details are displayed.	Trip details were displayed with trip ID, route, date, times and seat count.	Pass	Observed through the Trip menu.

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
TR-TC3	Travel date is stored correctly.	The travel date 2026-12-17 was stored and displayed correctly.	Pass	Date matched the input.
TR-TC4	Seat count is stored correctly.	The trip stored and displayed 4 total seats.	Pass	Seat capacity matched the input.
TR-TC5	Times are stored correctly.	Departure 08:00 and arrival 09:00 were stored and displayed correctly.	Pass	Both valid time values were retained.

Class Being Tested: TripManager

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
TM-TC1	Trip is created and stored.	The system printed Trip added successfully and saved the new trip.	Pass	Valid route, date, times and seats were accepted.
TM-TC2	Matching trip is returned.	Searching route 1 and date 2026-12-15 returned Trip 1.	Pass	Search returned the expected trip.
TM-TC3	Trip is updated successfully.	Trip 1 was updated to 2026-12-18, 10:00-11:00.	Pass	Saved trip data reflected the new values.
TM-TC4	Cancellation is prevented.	The system printed: Trip cannot be cancelled because it has active tickets.	Pass	Trip remained available.
TM-TC5	Seat map shows the current seat availability.	Seat layout displayed [1:O] [2:O] [3:O] [4:O].	Pass	All four seats were correctly shown as open.

Class Being Tested: Passenger

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
P-TC1	Passenger stores the supplied values correctly.	Passenger Emma Audit with phone 59900112 was created and saved.	Pass	Stored values matched the input.
P-TC2	Readable passenger details are displayed.	Passenger details were displayed with passenger ID, name and phone number.	Pass	Observed through display/search output.
P-TC3	Phone number is stored correctly.	Phone 59900112 was stored and displayed unchanged.	Pass	Phone remained a text value.
P-TC4	Full name is stored correctly.	Name Alice Audit was stored with the space preserved.	Pass	Full name remained intact.
P-TC5	Updated name is stored correctly.	Passenger 1 was successfully updated to Alice Updated.	Pass	Saved passenger data contained the updated name.

Class Being Tested: PassengerManager

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
PM-TC1	Passenger is created and stored.	Emma Audit was registered successfully and added to the saved passenger records.	Pass	Valid registration succeeded.
PM-TC2	Registration is rejected.	Empty name produced: Invalid name: name cannot be empty.	Pass	No passenger was created.
PM-TC3	Registration is rejected.	Numeric-only name 12345 was rejected after the validation fix.	Pass	Saved passenger records remained unchanged.

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
PM-TC4	Matching passenger is returned.	Searching passenger ID 1 returned the expected passenger.	Pass	Search matched the existing record.
PM-TC5	Update is rejected and the original passenger data remains unchanged.	Using a valid new name with phone WEW was rejected; neither field changed.	Pass	Re-tested after the atomic update fix.

Class Being Tested: BusPass

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
BP-TC1	BusPass stores the supplied values correctly.	A student pass for passenger 4 was created with the supplied dates and active status.	Pass	Saved values matched the input.
BP-TC2	Readable bus pass details are displayed.	The Bus Pass display showed pass ID, passenger ID, type, issue date, expiry date and status.	Pass	Observed through the Pass menu.
BP-TC3	Senior pass type is stored correctly.	A senior pass was created and stored as type senior.	Pass	Pass type matched the input.
BP-TC4	Priority pass type is stored correctly.	A priority pass was created and stored as type priority.	Pass	Pass type matched the input.
BP-TC5	Pass status changes to suspended.	Suspending pass 1 changed its status to suspended.	Pass	Saved pass data reflected the status change.

Class Being Tested: BusPassManager

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
BPM-TC1	Pass is created and stored.	A valid student pass was issued successfully.	Pass	Pass was added to the saved pass records.
BPM-TC2	Pass is not created.	Passenger 999 produced: Invalid passenger: passenger with ID 999 does not exist.	Pass	No pass was added.
BPM-TC3	Second active pass is rejected.	The system rejected issuing another active pass to a passenger who already had one.	Pass	Saved pass records remained unchanged.
BPM-TC4	Pass is renewed successfully.	Pass 1 renewed successfully with expiry date 2027-01-01.	Pass	Expiry date changed and the pass remained active.
BPM-TC5	Renewal is rejected so only one active pass remains.	Renewing the old suspended pass was rejected after a newer active pass had been issued.	Pass	Re-tested after the renewal fix; exactly one active pass remained.

Class Being Tested: Ticket

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
T-TC1	Ticket stores the supplied values correctly.	Ticket 1 was created for passenger 4, trip 1, seat 1, price 40.00, status active.	Pass	Stored ticket values matched the purchase.
T-TC2	Ticket is stored without a pass.	The ticket was created with Pass ID: None.	Pass	No bus pass was applied.

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
T-TC3	Ticket stores the applied pass and discounted price.	A student ticket was created with Pass ID 1 and price 20.00.	Pass	Student discount was correctly reflected in the stored ticket.
T-TC4	Ticket status changes to cancelled.	Cancelling ticket 1 changed its status to cancelled.	Pass	Cancelled ticket no longer occupied the seat.
T-TC5	Readable ticket details are displayed.	Ticket details were displayed with ticket ID, passenger, trip, seat, pass, price, purchase date and status.	Pass	Observed through display/search output.

Class Being Tested: TicketManager

Test ID	Expected Output	Actual Output	Pass/Fail	Comments
TKM-TC1	Ticket is created at the full route fare.	Normal ticket purchase succeeded at price 40.00 with no pass.	Pass	Full route fare was used.
TKM-TC2	Ticket is created at 50% of the route fare.	Valid student-pass purchase succeeded at price 20.00 from a 40.00 route fare.	Pass	Student fare calculation was correct.
TKM-TC3	Purchase is rejected.	Attempting to buy an already occupied seat produced: Seat 1 is already occupied.	Pass	No second active ticket was created for the seat.
TKM-TC4	Purchase is rejected.	Seat 5 on a 4-seat trip was rejected with a seat-range message.	Pass	No ticket was created.
TKM-TC5	Ticket is created at full fare without the pass benefit.	Using an expired pass created the ticket at full fare 40.00 with Pass ID None.	Pass	Expired pass did not reduce the fare.

